Week 7 Lab – Telling a Climate Story with One Chart

> **Goal for this lab:** Take *real* global temperature data and build up **one** clear, honest chart step by step:
>
> **start simple → add layers → add clarity → add truth → add narrative**

By the end, you should be able to:

- Load and tidy a real climate dataset from the web.
- Build a basic ggplot and then refine it layer by layer.
- Visualise **uncertainty** (not just the line).
- Add a **story focus** (e.g. "recent decades are very different from the past").

We will use global temperature anomaly data from the **Berkeley Earth** project – the same scientific basis behind many "warming stripes" style visuals.

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0. Setup

We'll use `tidyverse` and `readr` (already part of tidyverse).

```
library(tidyverse)
```

> If `tidyverse` is not installed in your session, run once (and only once, do not repeat during repeated runs):

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1. Load the Global Temperature Data

We'll use the **summary** file that contains annual global anomalies for land + ocean.

```
# URL from Berkeley Earth (Global Land + Ocean summary)
url <- "https://berkeley-earth-temperature.s3.us-west-1.amazonaws.com/Global/Land_and_Ocean_su

global_raw <- read_table(
  file      = url,
  comment   = "%",  # ignore documentation lines starting with %
  col_names = c(
    "year",
    "annual_anom_air",   "annual_unc_air",
    "fiveyr_anom_air",   "fiveyr_unc_air",
    "annual_anom_water", "annual_unc_water",
    "fiveyr_anom_water", "fiveyr_unc_water"
  )
)
```

```r
glimpse(global_raw)
```

```
Rows: 176
Columns: 9
$ year              <dbl> NA, 1850, 1851, 1852, 1853, 1854, 1855, 1856, 1857, …
$ annual_anom_air   <dbl> NA, -0.436, -0.321, -0.292, -0.295, -0.283, -0.278, …
$ annual_unc_air    <dbl> NA, 0.170, 0.259, 0.249, 0.196, 0.130, 0.120, 0.163,…
$ fiveyr_anom_air   <chr> NA, "NaN", "NaN", "-0.325", "-0.294", "-0.309", "-0.…
$ fiveyr_unc_air    <chr> NA, "NaN", "NaN", "0.173", "0.155", "0.137", "0.121"…
$ annual_anom_water <dbl> NA, -0.419, -0.319, -0.280, -0.285, -0.279, -0.263, …
$ annual_unc_water  <dbl> NA, 0.156, 0.225, 0.217, 0.173, 0.121, 0.112, 0.147,…
$ fiveyr_anom_water <chr> NA, "NaN", "NaN", "-0.316", "-0.285", "-0.299", "-0.…
$ fiveyr_unc_water  <chr> NA, "NaN", "NaN", "0.153", "0.138", "0.123", "0.111"…
```

What's in here?

- `annual_anom_air` – annual temperature anomaly (°C) relative to **1951–1980**, using **air** over sea ice
- `annual_unc_air` – 95% confidence interval width for that anomaly
- `fiveyr_anom_air` – smoothed 5-year anomaly (air)
- … and the same for water over sea ice (we'll ignore the water version for now)

Let's create a cleaned version focused on the **air** series and a reasonable time window (from 1850 onwards):

```r
global <- global_raw |>
  filter(year >= 1850) |>
  select(
    year,
    annual_anom_air,
    annual_unc_air,
    fiveyr_anom_air,
    fiveyr_unc_air
  )

summary(global)
```

```
     year      annual_anom_air     annual_unc_air    fiveyr_anom_air
 Min.   :1850   Min.   :-0.554000   Min.   :0.02600   Length:175
 1st Qu.:1894   1st Qu.:-0.295000   1st Qu.:0.03100   Class :character
 Median :1937   Median :-0.102000   Median :0.07800   Mode  :character
 Mean   :1937   Mean   : 0.008274   Mean   :0.07649
 3rd Qu.:1980   3rd Qu.: 0.194500   3rd Qu.:0.09050
 Max.   :2024   Max.   : 1.311000   Max.   :0.25900
 fiveyr_unc_air
 Length:175
 Class :character
 Mode  :character
```

Do you notice that two variables are encoded as text. It might be a good idea to convert these to numbers right here.

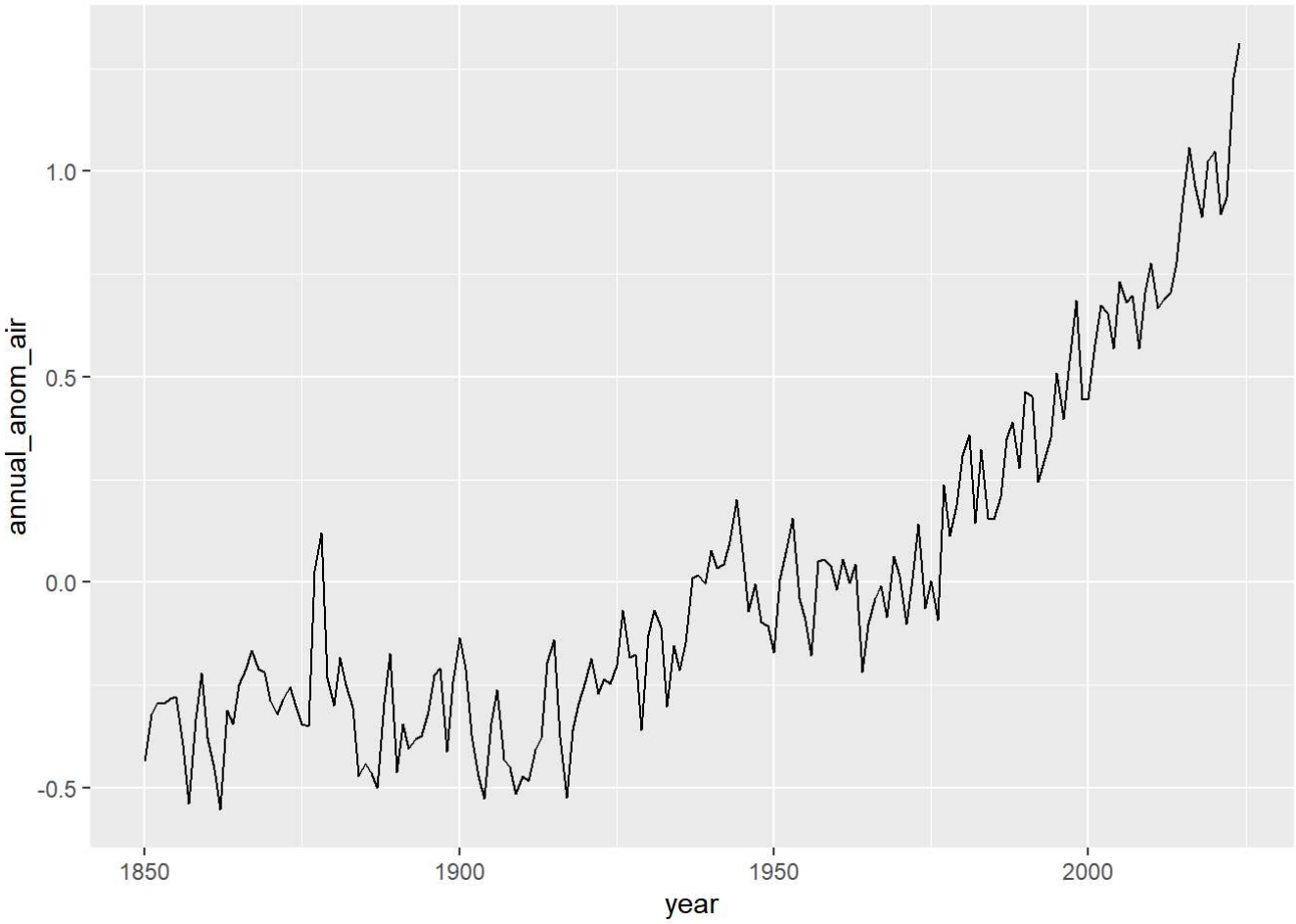
```
global <- global |>
  dplyr::mutate(
    annual_anom_air  = as.numeric(annual_anom_air),
    fiveyr_anom_air  = as.numeric(fiveyr_anom_air)
  )
```

---

2. Start Simple – One Line, No Frills

We begin with the most basic legitimate chart: a line of annual anomalies over time.

```
ggplot(global, aes(x = year, y = annual_anom_air)) +
  geom_line()
```



> **Think:**
>
> - Can you roughly see that the later years are warmer than the early years?
> - Is that it? What *can't* you see clearly yet?

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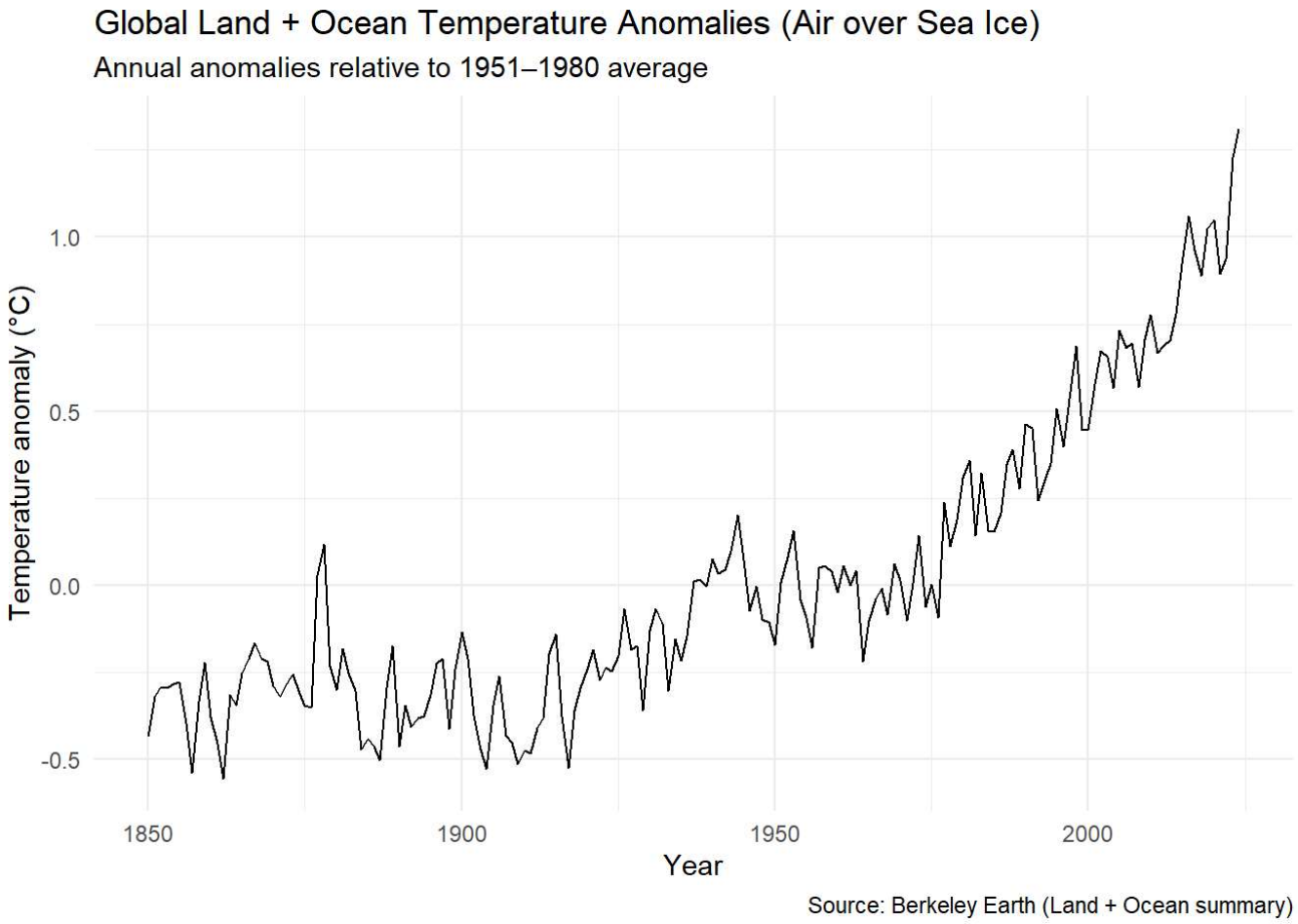
3. Add Layers – Make the Shape Legible

Now let's slowly improve readability without changing the data itself.

3.1 Add Labels and a Minimal Theme

```
ggplot(global, aes(x = year, y = annual_anom_air)) +
  geom_line() +
  labs(
    title = "Global Land + Ocean Temperature Anomalies (Air over Sea Ice)",
    subtitle = "Annual anomalies relative to 1951–1980 average",
    x = "Year",
    y = "Temperature anomaly (°C)",
    caption = "Source: Berkeley Earth (Land + Ocean summary)"
  ) +
  theme_minimal()
```



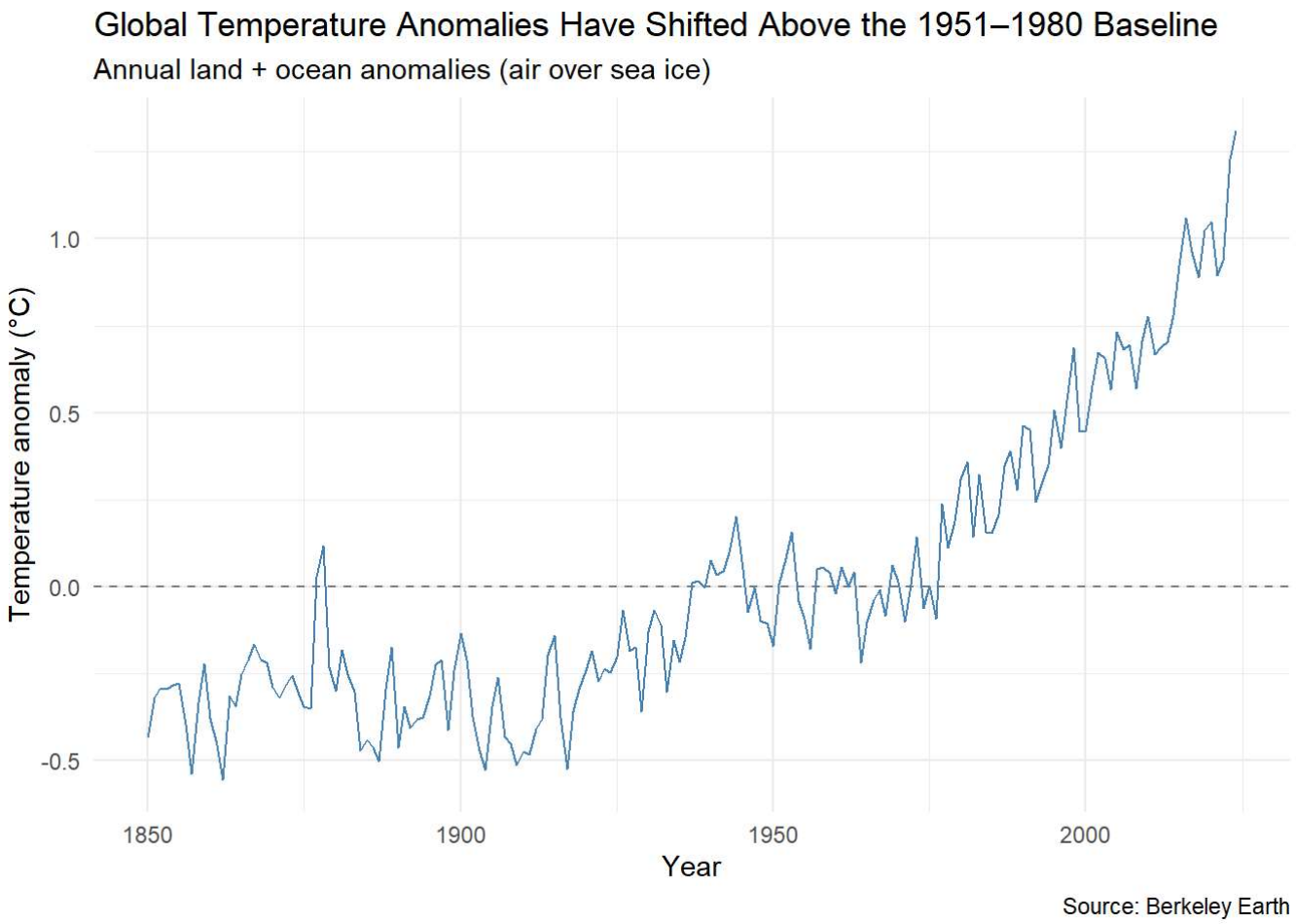
Already better: someone who has **never** heard of Berkeley Earth can now tell what this chart is about.

3.2 Add a Zero Reference Line (the "baseline")

We know anomalies are measured relative to 1951–1980. Let's make that baseline visible.

```
ggplot(global, aes(x = year, y = annual_anom_air)) +
  geom_hline(yintercept = 0, linetype = "dashed", colour = "grey50") +
  geom_line(colour = "steelblue") +
```

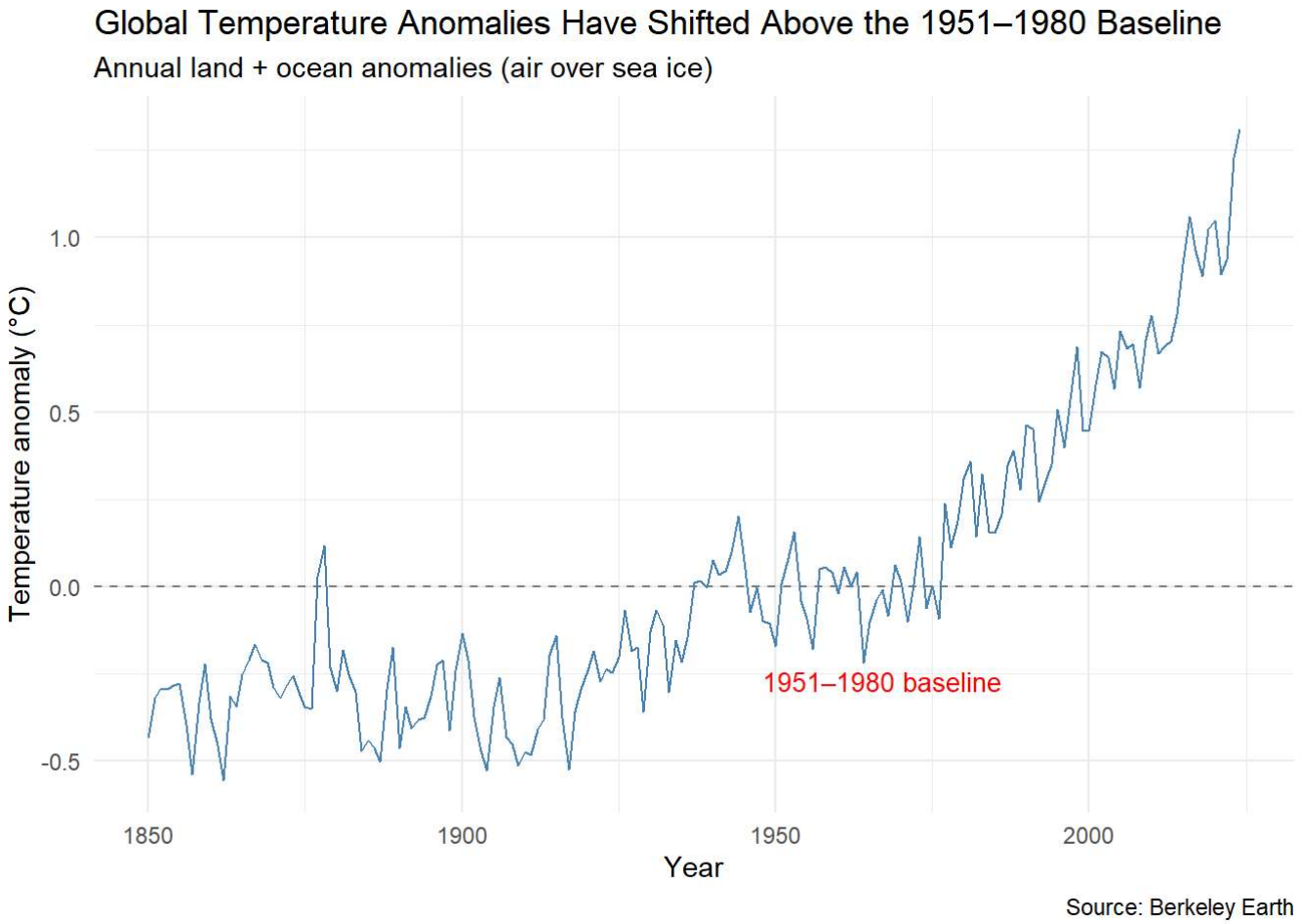
```
  labs(
    title = "Global Temperature Anomalies Have Shifted Above the 1951–1980 Baseline",
    subtitle = "Annual land + ocean anomalies (air over sea ice)",
    x = "Year",
    y = "Temperature anomaly (°C)",
    caption = "Source: Berkeley Earth"
  ) +
  theme_minimal()
```



3.3 Highlight the Baseline on Plot

```
ggplot(global, aes(x = year, y = annual_anom_air)) +
  geom_hline(
    yintercept = 0,
    linetype = "dashed",
    colour = "grey50"
  ) +
    # baseline label
  annotate(
    "text",
    x = 1948,          # left side so it does not overlap data
    y = -0.27,          # slightly above the baseline
    label = "1951–1980 baseline",
    hjust = 0,
    size = 3.5,
    colour = "red"
  ) +
```

```
  geom_line(colour = "steelblue") +
  labs(
    title = "Global Temperature Anomalies Have Shifted Above the 1951–1980 Baseline",
    subtitle = "Annual land + ocean anomalies (air over sea ice)",
    x = "Year",
    y = "Temperature anomaly (°C)",
    caption = "Source: Berkeley Earth"
  ) +
  theme_minimal()
```



> **Pause and interpret (in one sentence):** What does it mean that almost all recent years are **above** the dashed zero line?

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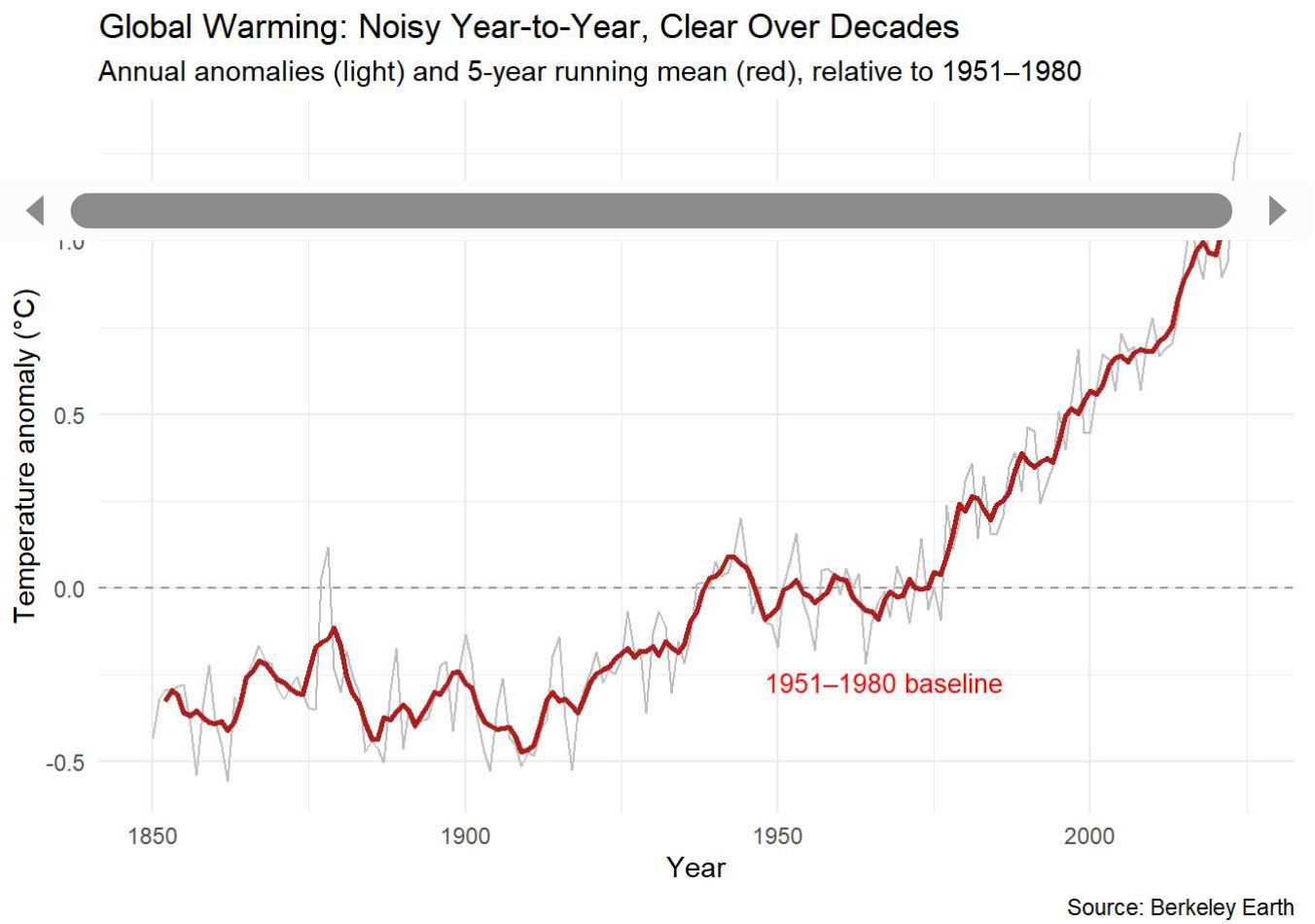
4. Add Clarity – Smooth the Noise (Without Lying)

Annual data can be noisy. Berkeley Earth also provides a **5-year smoothed anomaly**. Let's overlay it.

4.1 Two Lines: Annual vs 5-Year Smoothed

```
ggplot(global, aes(x = year)) +
  geom_hline(yintercept = 0, linetype = "dashed", colour = "grey60") +
  geom_line(aes(y = annual_anom_air),
            colour = "grey75",
            linewidth = 0.4) +
  geom_line(aes(y = fiveyr_anom_air),
```

```
            colour = "firebrick",
            linewidth = 1) +
  # baseline label
annotate(
  "text",
  x = 1948,          # left side so it does not overlap data
  y = -0.27,          # slightly above the baseline
  label = "1951–1980 baseline",
  hjust = 0,
  size = 3.5,
  colour = "red"
) +
labs(
  title = "Global Warming: Noisy Year-to-Year, Clear Over Decades",
  subtitle = "Annual anomalies (light) and 5-year running mean (red), relative to 1951–1980"
  x = "Year",
  y = "Temperature anomaly (°C)",
  caption = "Source: Berkeley Earth"
) +
theme_minimal()
```



- Grey thin line: shows **annual ups and downs**
- Red thick line: emphasises the **decadal trend**

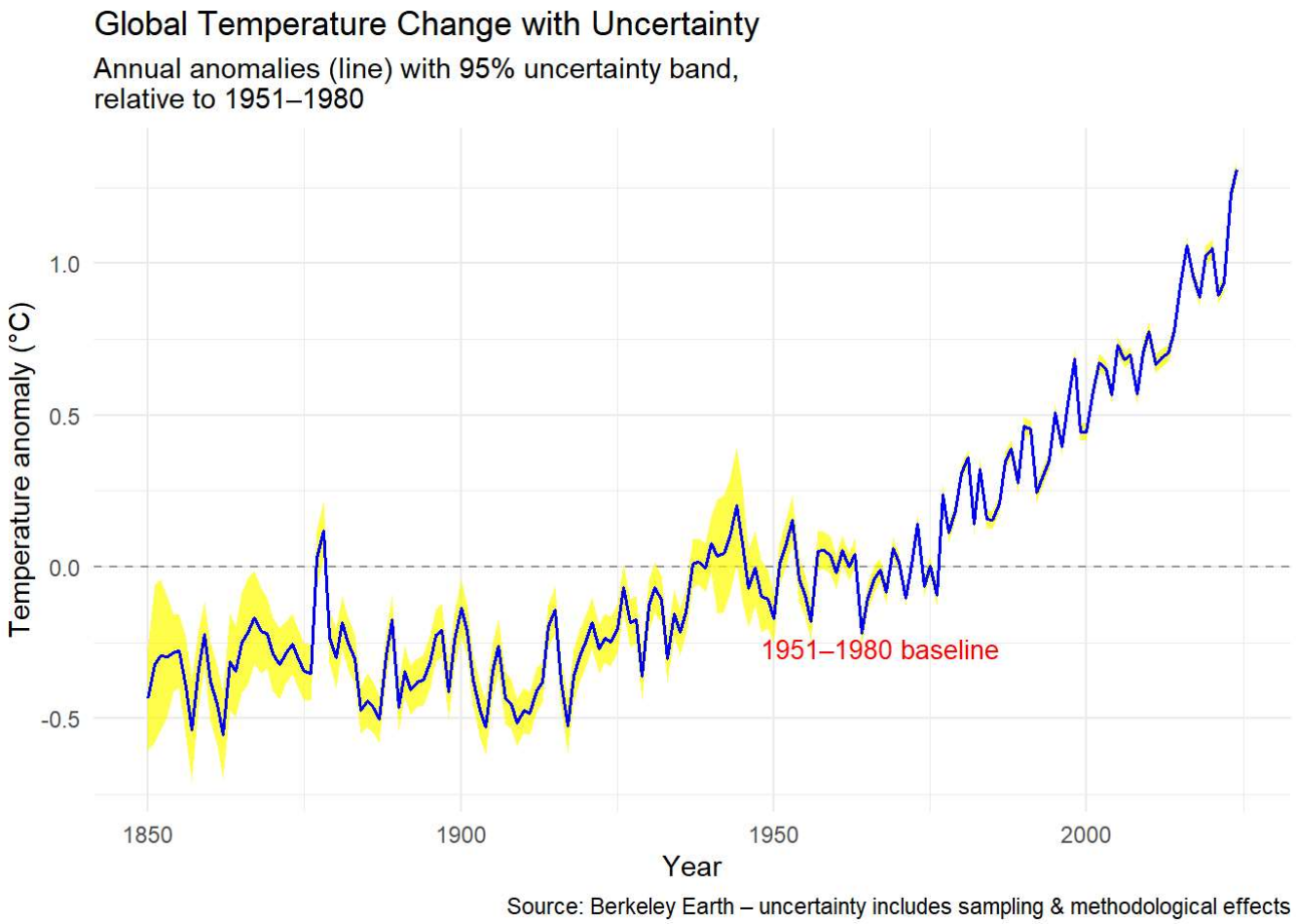
> **Reflect:** Is this smoothing helping or hiding something? When might smoothing be *misleading*?

5. Add Truth – Show Uncertainty, Not Just the Line

So far, our lines look precise. In reality, each estimate has **uncertainty**.

We'll add a ribbon showing the 95% confidence range around the **annual anomaly**.

```
ggplot(global, aes(x = year, y = annual_anom_air)) +
  geom_hline(yintercept = 0, linetype = "dashed", colour = "grey60") +
  geom_ribbon(
    aes(
      ymin = annual_anom_air - annual_unc_air,
      ymax = annual_anom_air + annual_unc_air
    ),
    fill = "yellow",
    alpha = 0.7
  ) +
    # baseline label
  annotate(
    "text",
    x = 1948,         # left side so it does not overlap data
    y = -0.27,         # slightly above the baseline
    label = "1951–1980 baseline",
    hjust = 0,
    size = 3.5,
    colour = "red"
  ) +
  geom_line(colour = "blue", linewidth = 0.7) +
  labs(
    title = "Global Temperature Change with Uncertainty",
    subtitle = "Annual anomalies (line) with 95% uncertainty band,\nrelative to 1951–1980",
    x = "Year",
    y = "Temperature anomaly (°C)",
    caption = "Source: Berkeley Earth – uncertainty includes sampling & methodological effects
  ) +
  theme_minimal()
```



> **Think:**
>
> - When is the ribbon wider? When is it narrower?
> - Why might estimates in early years be less certain?

---

6. Add Narrative – Focus on a Story You Care About

Now we have a technically honest plot (trend + baseline + uncertainty). Next step: pick a **story** and shape the chart around it.

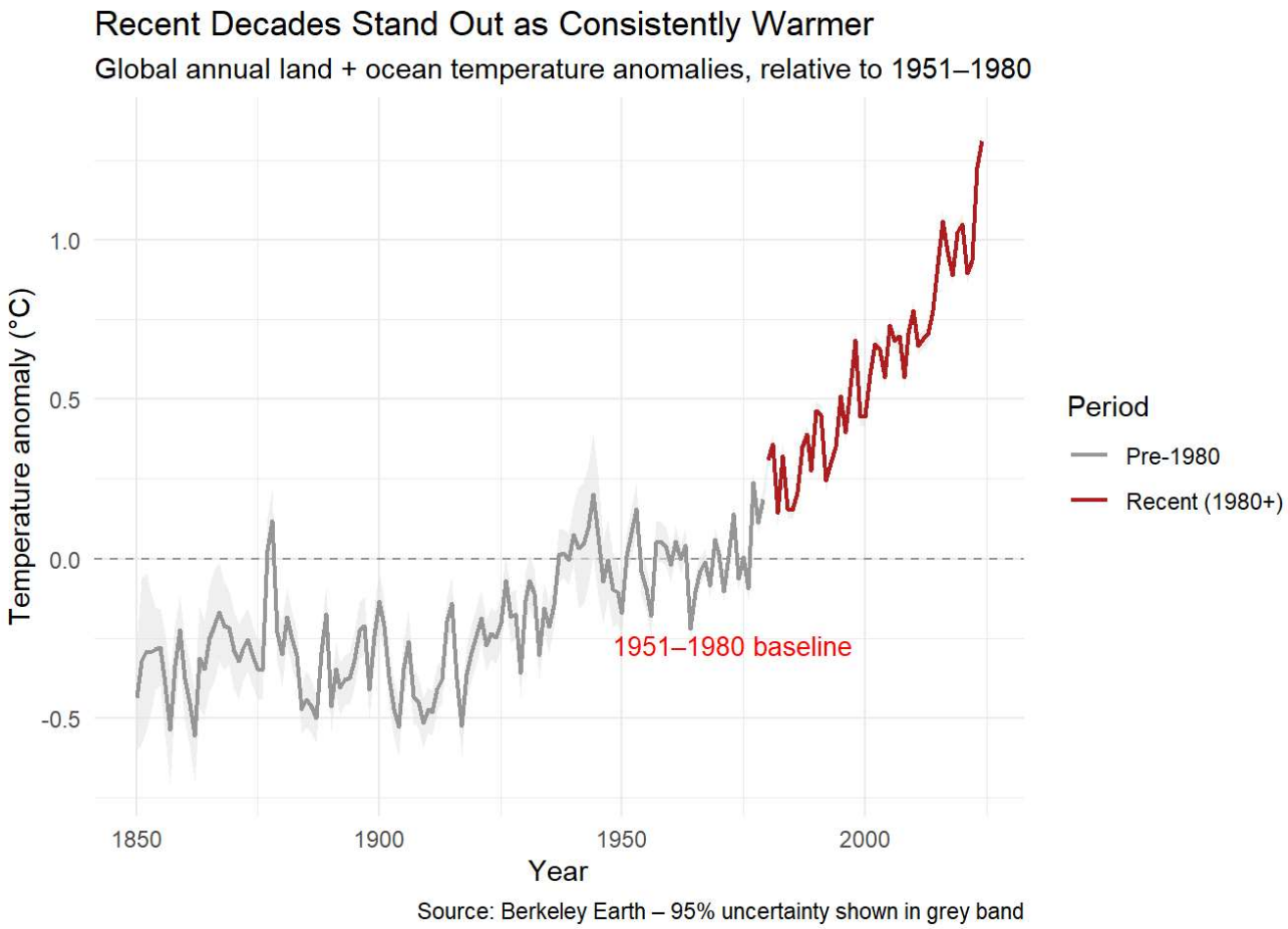
We'll build one concrete example:

> **Story:** "Recent decades are markedly warmer than any previous period since 1850."

6.1 Highlight the Recent Decades (e.g. 1980 onwards)

```
global <- global |>
  mutate(
    period = if_else(year >= 1980, "Recent (1980+)", "Pre-1980")
  )

ggplot(global, aes(x = year, y = annual_anom_air)) +
  geom_hline(yintercept = 0, linetype = "dashed", colour = "grey60") +
```

```r
geom_ribbon(
  aes(
    ymin = annual_anom_air - annual_unc_air,
    ymax = annual_anom_air + annual_unc_air
  ),
  fill = "grey90",
  alpha = 0.5
) +
  # baseline label
annotate(
  "text",
  x = 1948,         # left side so it does not overlap data
  y = -0.27,         # slightly above the baseline
  label = "1951–1980 baseline",
  hjust = 0,
  size = 3.5,
  colour = "red"
) +
geom_line(
  aes(colour = period),
  linewidth = 0.8
) +
scale_colour_manual(
  values = c("Pre-1980" = "grey60", "Recent (1980+)" = "firebrick")
) +
labs(
  title = "Recent Decades Stand Out as Consistently Warmer",
  subtitle = "Global annual land + ocean temperature anomalies, relative to 1951–1980",
  x = "Year",
  y = "Temperature anomaly (°C)",
  colour = "Period",
  caption = "Source: Berkeley Earth – 95% uncertainty shown in grey band"
) +
theme_minimal()
```



Now the **visual emphasis matches the message**: we're telling a story about the recent period, so the colour focus is there.

6.2 Annotate the Hottest Years

Let's label the **top 5 hottest years** to give the reader "anchors".

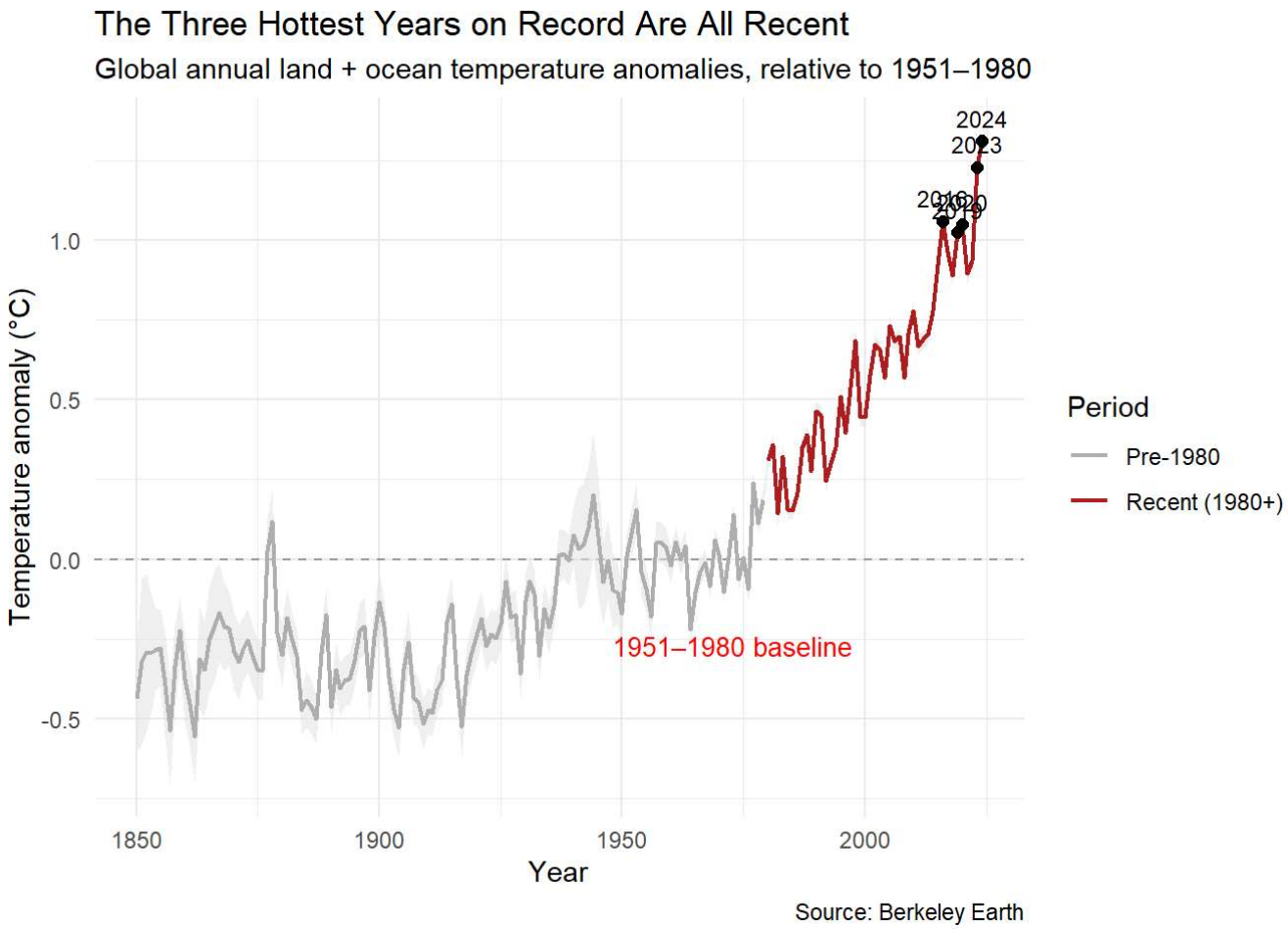
```
top_years <- global |>
  arrange(desc(annual_anom_air)) |>
  slice_head(n = 5)

top_years
```

```
# A tibble: 5 × 6
   year annual_anom_air annual_unc_air fiveyr_anom_air fiveyr_unc_air period
  <dbl>           <dbl>          <dbl>           <dbl> <chr>          <chr>
1  2024            1.31          0.032         NaN     NaN            Recent (1…
2  2023            1.23          0.031         NaN     NaN            Recent (1…
3  2016            1.06          0.027           0.922 0.024          Recent (1…
4  2020            1.05          0.032           0.958 0.026          Recent (1…
5  2019            1.02          0.031           0.964 0.025          Recent (1…
```

Now annotate them:

```
ggplot(global, aes(x = year, y = annual_anom_air)) +
  geom_hline(yintercept = 0, linetype = "dashed", colour = "grey60") +
  geom_ribbon(
```

```r
    aes(
      ymin = annual_anom_air - annual_unc_air,
      ymax = annual_anom_air + annual_unc_air
    ),
    fill = "grey90",
    alpha = 0.5
  ) +
  geom_line(
    aes(colour = period),
    linewidth = 0.8
  ) +
    # baseline label
  annotate(
    "text",
    x = 1948,         # left side so it does not overlap data
    y = -0.27,         # slightly above the baseline
    label = "1951–1980 baseline",
    hjust = 0,
    size = 3.5,
    colour = "red"
  ) +
  geom_point(data = top_years, colour = "black", size = 2) +
  geom_text(
    data = top_years,
    aes(label = year),
    vjust = -0.8,
    size = 3
  ) +
  scale_colour_manual(
    values = c("Pre-1980" = "grey70", "Recent (1980+)" = "firebrick")
  ) +
  labs(
    title = "The Three Hottest Years on Record Are All Recent",
    subtitle = "Global annual land + ocean temperature anomalies, relative to 1951–1980",
    x = "Year",
    y = "Temperature anomaly (°C)",
    colour = "Period",
    caption = "Source: Berkeley Earth"
  ) +
  theme_minimal()
```



> **Notice:** We haven't changed the data — just improved focus, context, and interpretability.

---

7. Your Turn

Now it's your turn to apply the **start simple → add layers → add clarity → add truth → add narrative** pattern.

You may use [this version]